

Is Australia Doing Well? Stress-Testing ‘It’s Green Again’

A conservative test of whether post-fire vegetation greenness is a reliable proxy for wildlife recovery under heat exposure

[Team members + SID]

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1 1. Research question

Is Australia’s fire-affected environment doing well?

The honest answer this analysis lands on is: **it depends on which layer of “the environment” you look at, and the evidence is not strong enough to treat vegetation greenness as a complete recovery signal.** The central result after the stricter method fixes in this report is this:

Under a conservative recovery-ratio specification, we do not find robust evidence that post-fire heat exposure widens the gap between vegetation recovery and wildlife-sighting recovery (Spearman $\rho = 0.001$, $p = 0.996$; all model-based checks are also non-significant). This does not prove that NDVI is a complete habitat-recovery measure. It shows that the stronger “heat widens the blind spot” claim is not stable once the heat window is aligned to the recovery window and heat exposure is measured against a documented 1991-2020 local normal.

Everything below builds and stress-tests that claim: the dataset, the event-level construction, an interactive map, the statistical checks, and a direct answer to the research question.

2 2. Setup

```

import os
import json
import numpy as np
import pandas as pd
from scipy import stats
import matplotlib.pyplot as plt
import plotly.express as px

DATA_DIR = os.path.join(os.getcwd(), "..", "data") # report lives in reports/, data lives in
pd.set_option("display.width", 140)

def cluster_robust_ols(y, X, clusters, names):
    """OLS with one-way cluster-robust standard errors, avoiding non-core dependencies."""
    y = np.asarray(y, dtype=float)
    X = np.asarray(X, dtype=float)
    clusters = np.asarray(clusters)
    n, k = X.shape
    xtx_inv = np.linalg.pinv(X.T @ X)
    beta = xtx_inv @ X.T @ y
    resid = y - X @ beta

    meat = np.zeros((k, k))
    unique_clusters = np.unique(clusters)
    for cluster in unique_clusters:
        idx = clusters == cluster
        xg = X[idx]
        ug = resid[idx][:, None]
        meat += xg.T @ ug @ ug.T @ xg

    g = len(unique_clusters)
    correction = (g / (g - 1)) * ((n - 1) / (n - k))
    cov = correction * xtx_inv @ meat @ xtx_inv
    se = np.sqrt(np.diag(cov))
    t_stat = beta / se
    p_value = 2 * stats.t.sf(np.abs(t_stat), df=g - 1)
    return pd.DataFrame({
        "term": names,
        "estimate": beta,
        "std_error_cluster": se,
        "t_stat": t_stat,
        "p_value": p_value,
        "n": n,
    })

```

```
"clusters": g,
})
```

3 3. Data and event-level construction

Four datasets, all already built/cached by the companion notebooks in this folder (`fire_history_edu.ipynb`, `fire_heat_recovery_analysis.qmd`):

Dataset	Role
Multi-state fire history shapefiles	major fire events ($\geq 10,000$ ha) and severity, per LGA-division-year
AusENDVI-clim_MCD43A4...nc	vegetation recovery signal (NDVI)
animal_sightings.csv	koala / greater glider / yellow-bellied glider presence
ERA5-Land daily max temperature	post-fire heat exposure, relative to each division's 1991-2020 hot-day normal

Koalas and gliders are used as targeted **forest-dependent arboreal indicator species**, not as a random sample of animals. The sharper ecological logic is for gliders: greater and yellow-bellied gliders rely on mature forest structure, hollow-bearing trees, den sites, food resources and canopy connectivity – habitat features that can remain degraded even when satellite NDVI shows that vegetation is green again. Koalas are retained as an iconic forest-dependent comparison taxon with many records. The animal data are still presence records rather than abundance estimates, so they are interpreted as a wildlife recovery signal, not a direct population measure. This species choice is supported by Australian post-fire recovery priorities for greater and yellow-bellied gliders, literature on hollow-bearing tree dependence, and recovery-plan documentation that identifies changed fire regimes and loss of key habitat trees as threats (see References).

The heat metric follows documented climate-index logic in two steps. First, annual hot days are counted using a fixed 35C daily maximum temperature threshold, consistent with the Bureau of Meteorology hot-day index and Climdex TXge35. Second, the event-window value is expressed as an anomaly: observed hot-day count minus a local long-term normal. The 1991-2020 baseline uses World Meteorological Organization climatological-normal practice, while the event-window averaging is study-specific and aligned to the recovery assessment window.

The analysis unit is the **individual major fire event** (division x fire-year, fire $\geq 10,000$ ha) – not the whole division. Vegetation and wildlife recovery are both computed as a Relative Recovery Indicator (RRI, adapted from Frazier et al., 2018 – see Section 11), which requires a measurable dip in the fire year itself; that requirement is easy for NDVI (166 of 184 candidate

events keep a valid ratio) but more demanding for sparse citizen-science sightings data (49 events, across 27 divisions, keep a valid ratio) – see Section 9 for what that does to statistical power.

```

ndvi = pd.read_csv(os.path.join(DATA_DIR, "ndvi_annual_by_division.csv")).set_index(["ADM1_NAME", "ADM2_NAME"])
ndvi.columns = ndvi.columns.astype(int)

fire_events = pd.read_csv(os.path.join(DATA_DIR, "fire_events_by_division_year.csv"))
major = fire_events[fire_events.major_fires > 0].copy()

sightings = pd.read_csv(os.path.join(DATA_DIR, "animal_sightings_by_division_year.csv"))
div_year = sightings.groupby(["ADM1_NAME", "ADM2_NAME", "year"])["sightings"].sum().reset_index()
total_per_year = div_year.groupby("year")["sightings"].sum().rename("total_sightings_all_divisions")
div_year = div_year.merge(total_per_year, on="year")
div_year["share"] = div_year["sightings"] / div_year["total_sightings_all_divisions"]
share_piv = div_year.pivot_table(index=["ADM1_NAME", "ADM2_NAME"], columns="year", values="share")
total_sightings_per_div = sightings.groupby(["ADM1_NAME", "ADM2_NAME"])["sightings"].sum()

heat = pd.read_csv(os.path.join(DATA_DIR, "heat_stress_by_division_year.csv"))
heat_piv = heat.pivot_table(index=["ADM1_NAME", "ADM2_NAME"], columns="year", values="hot_days")
HEAT_BASELINE_YEARS = [y for y in range(1991, 2021) if y in heat_piv.columns]
heat_baseline_mean = heat_piv[HEAT_BASELINE_YEARS].mean(axis=1)

PRE_YEARS, RECOVERY_POST_LAGS, SMOOTH = 2, [3, 4, 5], 1e-6

records = []
for _, r in major.iterrows():
    state, div, year = r["ADM1_NAME"], r["ADM2_NAME"], int(r["year"])
    key = (state, div)
    if key not in ndvi.index or key not in heat_piv.index:
        continue
    ndvi_row, heat_row = ndvi.loc[key], heat_piv.loc[key]
    pre_years = [year - k for k in range(1, PRE_YEARS + 1)]
    recovery_post_years = [year + k for k in RECOVERY_POST_LAGS]

    # Relative Recovery Indicator (RRI), adapted from Frazier et al. (2018):
    # RRI = ARI / disturbance magnitude. Disturbance magnitude = pre-fire
    # baseline minus the fire-year (Y0) value; ARI = best post-fire value
    # observed minus Y0. The paper uses a fixed Y+4/Y+5 window, but NDVI here
    # ends in 2022, so a fixed Y+4/Y+5 window would drop every fire from 2019
    # onward (including the Black Summer cohort); instead we take the best
    # available of Y+3/Y+4/Y+5.
    pre_ndvi = [v for v in (ndvi_row.get(y) for y in pre_years) if pd.notna(v)]

```

```

y0_ndvi = ndvi_row.get(year)
available_recovery_years = [y for y in recovery_post_years if pd.notna(ndvi_row.get(y))]
post_ndvi = [ndvi_row.get(y) for y in available_recovery_years]
if not pre_ndvi or pd.isna(y0_ndvi) or not post_ndvi:
    continue
ndvi_baseline = np.mean(pre_ndvi)
ndvi_disturbance = ndvi_baseline - y0_ndvi
if ndvi_disturbance <= 0:
    continue
ndvi_ratio = (max(post_ndvi) - y0_ndvi) / ndvi_disturbance

sight_ratio = np.nan
if key in share_piv.index:
    srow = share_piv.loc[key]
    pre_share = [v for v in (srow.get(y) for y in pre_years) if pd.notna(v)]
    y0_share = srow.get(year)
    post_share = [v for v in (srow.get(y) for y in available_recovery_years) if pd.notna(v)]
    if pre_share and pd.notna(y0_share) and post_share:
        share_baseline = np.mean(pre_share) + SMOOTH
        y0_share_s = y0_share + SMOOTH
        share_disturbance = share_baseline - y0_share_s
        if share_disturbance > 0:
            sight_ratio = (max(post_share) + SMOOTH - y0_share_s) / share_disturbance

post_heat = [v for v in (heat_row.get(y) for y in available_recovery_years) if pd.notna(v)]
if not post_heat:
    continue
heat_post_mean = np.mean(post_heat)
heat_anomaly = heat_post_mean - heat_baseline_mean.get(key, np.nan)

records.append({
    "ADM1_NAME": state, "ADM2_NAME": div, "fire_year": year,
    "max_area_ha": r["max_area_ha"], "major_fires": r["major_fires"],
    "ndvi_recovery_ratio": ndvi_ratio, "sightings_recovery_ratio": sight_ratio,
    "heat_anomaly_postfire": heat_anomaly,
    "division_total_sightings": total_sightings_per_div.get(key, 0),
})

events_all = pd.DataFrame(records)
# RRI can be zero/negative (best post-fire value never exceeds the fire-year
# value); those events can't be log-transformed and are dropped.
events_all = events_all[events_all["ndvi_recovery_ratio"] > 0].copy()

```

```

events_all["log_ndvi_ratio"] = np.log(events_all["ndvi_recovery_ratio"])
events_all["log_severity"] = np.log(events_all["max_area_ha"])

HABITAT_MIN_SIGHTINGS = 200
events = events_all[(events_all["division_total_sightings"] >= HABITAT_MIN_SIGHTINGS)
                    & events_all["sightings_recovery_ratio"].notna()
                    & (events_all["sightings_recovery_ratio"] > 0)].copy()
events["log_sight_ratio"] = np.log(events["sightings_recovery_ratio"])
events["recovery_gap"] = events["log_ndvi_ratio"] - events["log_sight_ratio"]

print(f"{len(events_all)} major-fire events total; {len(events)} events with a valid recovery
      f"gap across {events['ADM2_NAME'].nunique()} divisions with real koala/glider presence
      f"(>= {HABITAT_MIN_SIGHTINGS} sightings on record)")

```

400 major-fire events total; 49 events with a valid recovery gap across 27 divisions with real

NDVI/sightings recovery ratio = RRI = (best post-fire value observed - fire-year value) / (pre-fire baseline - fire-year value) – i.e. how much of the fire-year dip was recovered, not a plain year+3/baseline ratio. **Recovery gap** = log(NDVI recovery ratio) - log(sightings recovery ratio). Zero means vegetation and wildlife recovered by the same proportional amount relative to how hard each was hit; positive means vegetation recovered *relatively better* than wildlife did.

4 4. EDA

```

events[["heat_anomaly_postfire", "ndvi_recovery_ratio", "sightings_recovery_ratio", "recovery_gap"]]

```

	heat_anomaly_postfire	ndvi_recovery_ratio	sightings_recovery_ratio	recovery_gap
count	49.000000	49.000000	49.000000	49.000000
mean	-1.242770	4.366376	11.003391	0.114583
std	6.390037	10.563856	27.777266	2.382635
min	-17.973588	0.079690	0.013924	-3.646718
25%	-3.518116	1.220249	0.606296	-1.717235
50%	-1.400570	2.180705	1.389549	-0.081295
75%	0.675489	3.624377	7.553657	1.505105
max	21.852113	74.078020	162.002857	6.031864

```

fig, axes = plt.subplots(1, 3, figsize=(15, 4))

axes[0].hist(events["heat_anomaly_postfire"], bins=30, color="steelblue")
axes[0].set_title("Post-Fire Heat Anomaly")
axes[0].set_xlabel("Extra hot days/yr (>=35C) vs. division's own baseline")
axes[0].set_ylabel("Number of fire events")

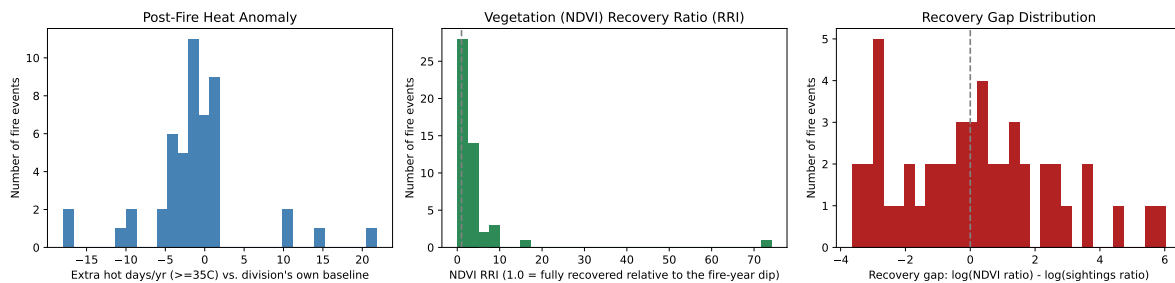
axes[1].hist(events["ndvi_recovery_ratio"], bins=30, color="seagreen")
axes[1].axvline(1.0, color="grey", ls="--")
axes[1].set_title("Vegetation (NDVI) Recovery Ratio (RRI)")
axes[1].set_xlabel("NDVI RRI (1.0 = fully recovered relative to the fire-year dip)")
axes[1].set_ylabel("Number of fire events")

axes[2].hist(events["recovery_gap"], bins=30, color="firebrick")
axes[2].axvline(0, color="grey", ls="--")
axes[2].set_title("Recovery Gap Distribution")
axes[2].set_xlabel("Recovery gap: log(NDVI ratio) - log(sightings ratio)")
axes[2].set_ylabel("Number of fire events")

fig.suptitle("Figure 1. Distributions of heat exposure, vegetation recovery, and the recovery gap (49 major fire events)", y=1.04)
plt.tight_layout()
plt.show()

```

Figure 1. Distributions of heat exposure, vegetation recovery, and the recovery gap (49 major fire events)



The gap distribution is centred close to zero with a long right tail. Some events show vegetation recovering much better than wildlife, but the key question is whether heat exposure reliably predicts which side of zero an event lands on.

5 5. Interactive map: where does the gap show up?

Each fire-prone division's **mean recovery gap** across all its major-fire events, 1982-2022. Red = vegetation recovers substantially better than wildlife (the “greenwashing risk” divisions); blue = the two track each other. Hover any division for its heat exposure and both recovery ratios.

```
div_agg = events.groupby(["ADM1_NAME", "ADM2_NAME"]).agg(
    n_events=("recovery_gap", "size"),
    mean_recovery_gap=("recovery_gap", "mean"),
    mean_heat_anomaly=("heat_anomaly_postfire", "mean"),
    mean_ndvi_ratio=("ndvi_recovery_ratio", "mean"),
    mean_sightings_ratio=("sightings_recovery_ratio", "mean"),
).reset_index()

with open(os.path.join(DATA_DIR, "au_admin2_boundaries.geojson"), "r") as f:
    admin2_geojson = json.load(f)

divisions_to_plot = set(div_agg["ADM2_NAME"])
geojson = {
    "type": "FeatureCollection",
    "features": [
        feature for feature in admin2_geojson["features"]
        if feature["properties"].get("ADM2_NAME") in divisions_to_plot
    ],
}

div_agg_labelled = div_agg.rename(columns={
    "ADM1_NAME": "State",
    "mean_recovery_gap": "Mean recovery gap",
    "mean_heat_anomaly": "Mean post-fire heat anomaly (extra hot days/yr)",
    "mean_ndvi_ratio": "Mean NDVI recovery ratio",
    "mean_sightings_ratio": "Mean sightings recovery ratio",
    "n_events": "Number of major fire events",
})

fig_map = px.choropleth_map(
    div_agg_labelled,
    geojson=geojson,
    locations="ADM2_NAME",
    featureidkey="properties.ADM2_NAME",
    color="Mean recovery gap",
```



```

color_continuous_scale="RdBu_r",
color_continuous_midpoint=0,
hover_name="ADM2_NAME",
hover_data={
    "State": True,
    "Mean recovery gap": ":.2f",
    "Mean post-fire heat anomaly (extra hot days/yr)": ":.1f",
    "Mean NDVI recovery ratio": ":.2f",
    "Mean sightings recovery ratio": ":.2f",
    "Number of major fire events": True,
},
map_style="carto-positron",
zoom=3.2,
center={"lat": -28, "lon": 145},
opacity=0.75,
height=650,
)
fig_map.update_layout(
    title=dict(
        text="Figure 2. Mean Post-Fire Recovery Gap by Fire-Prone Division<br>"
        "<sup>Red = vegetation recovers better than wildlife; Blue = the two track together</sup>"
        x=0.02, y=0.98,
    ),
    margin=dict(l=0, r=0, t=70, b=0),
    coloraxis_colorbar=dict(title="Mean recovery gap<br>(+ = vegetation<br>outpaces wildlife)"
)
fig_map

```

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The map is useful for seeing where the largest gaps occur, but it should not be read as proof of a heat effect. The statistical checks below test that relationship directly.

6 6. Visual evidence

6.1 6.1 Diverging recovery lines

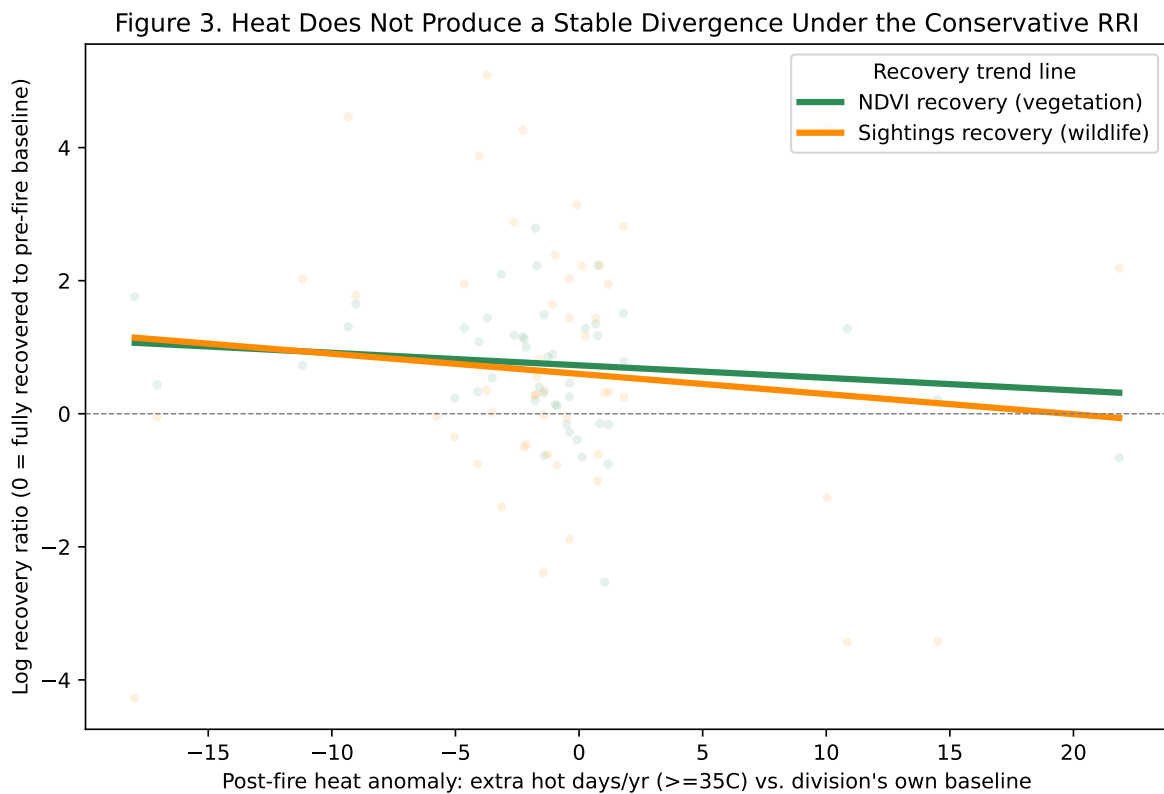
The single clearest picture of the claim: plot both recovery ratios (log scale, 0 = back to baseline) against post-fire heat exposure and fit a trend line to each.

```

fig, ax = plt.subplots(figsize=(8, 5.5))
xs = np.linspace(events["heat_anomaly_postfire"].min(), events["heat_anomaly_postfire"].max(), 100)
z_ndvi = np.polyfit(events["heat_anomaly_postfire"], events["log_ndvi_ratio"], 1)
z_sight = np.polyfit(events["heat_anomaly_postfire"], events["log_sight_ratio"], 1)

ax.scatter(events["heat_anomaly_postfire"], events["log_ndvi_ratio"], alpha=0.12, s=12, color="darkgreen")
ax.scatter(events["heat_anomaly_postfire"], events["log_sight_ratio"], alpha=0.12, s=12, color="darkorange")
ax.plot(xs, np.polyval(z_ndvi, xs), color="seagreen", lw=3, label="NDVI recovery (vegetation)")
ax.plot(xs, np.polyval(z_sight, xs), color="darkorange", lw=3, label="Sightings recovery (wildlife)")
ax.axhline(0, color="grey", lw=0.7, ls="--")
ax.set_xlabel("Post-fire heat anomaly: extra hot days/yr (>=35C) vs. division's own baseline")
ax.set_ylabel("Log recovery ratio (0 = fully recovered to pre-fire baseline)")
ax.set_title("Figure 3. Heat Does Not Produce a Stable Divergence Under the Conservative RRI")
ax.legend(title="Recovery trend line", loc="upper right")
plt.tight_layout()
plt.show()

```

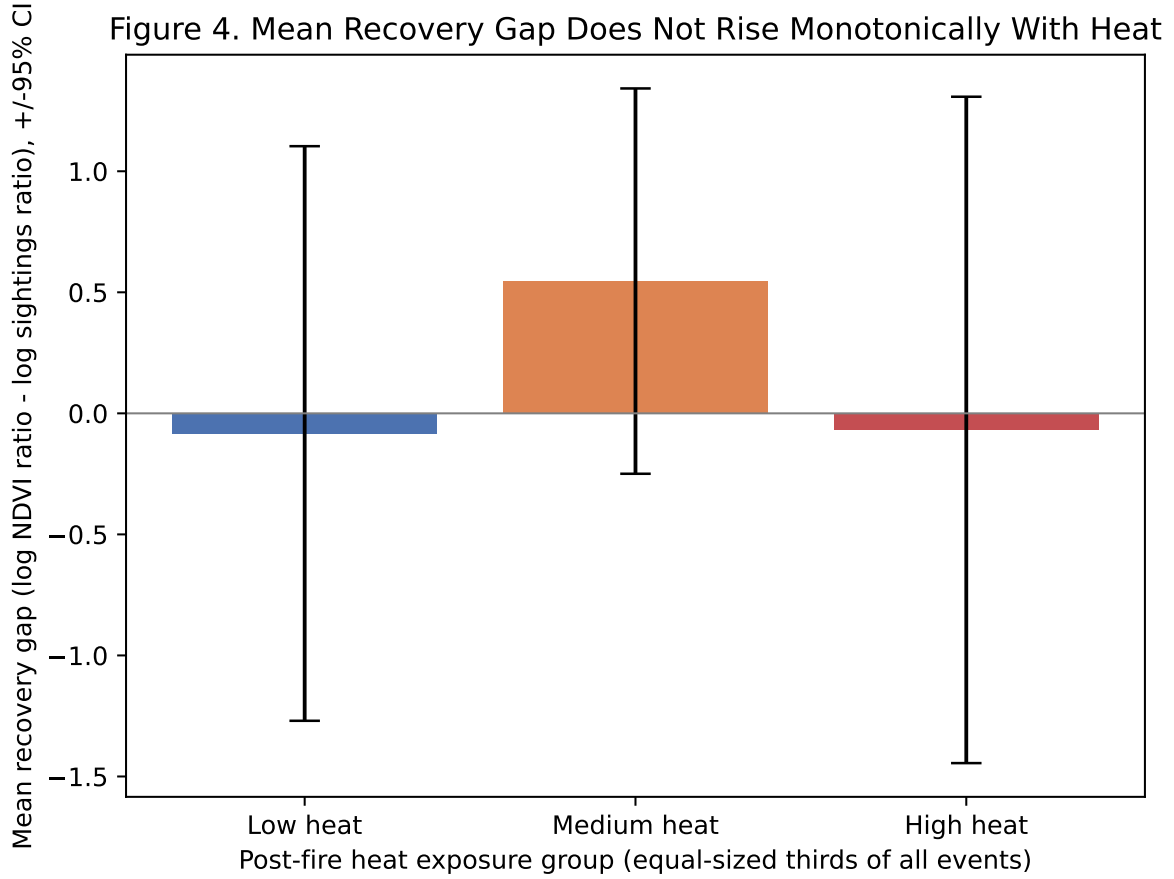


Under the conservative specification, the vegetation and wildlife trend lines do not separate in a stable way as heat exposure rises. The visual evidence is therefore weaker than the original

hypothesis predicted.

6.2 6.2 The gap by heat tercile

```
events["heat_tercile"] = pd.qcut(events["heat_anomaly_postfire"], 3, labels=["Low heat", "Me
tercile = events.groupby("heat_tercile", observed=True).agg(
    mean_gap=("recovery_gap", "mean"),
    se_gap=("recovery_gap", lambda x: x.std() / np.sqrt(len(x))),
)
fig, ax = plt.subplots(figsize=(6.5, 5))
ax.bar(tercile.index, tercile["mean_gap"], yerr=1.96 * tercile["se_gap"], capsize=6,
      color=["#4c72b0", "#dd8452", "#c44e52"])
ax.axhline(0, color="grey", lw=0.8)
ax.set_xlabel("Post-fire heat exposure group (equal-sized thirds of all events)")
ax.set_ylabel("Mean recovery gap (log NDVI ratio - log sightings ratio), +/-95% CI")
ax.set_title("Figure 4. Mean Recovery Gap Does Not Rise Monotonically With Heat")
plt.tight_layout()
plt.show()
tercile
```



Splitting events into thirds by heat exposure gives no monotonic pattern. The middle third has the largest mean gap, while the coolest and hottest thirds are both close to zero on average.

7 7. Statistical testing

Five complementary checks are used so the conclusion does not rest on one modelling choice.

7.1 7.1 Rank correlation

```
r, p = stats.spearmanr(events["heat_anomaly_postfire"], events["recovery_gap"])
print(f"Spearman(heat anomaly, recovery gap): rho={r:.3f}, p={p:.4g}, n={len(events)}")
```

Spearman(heat anomaly, recovery gap): rho=0.001, p=0.9956, n=49

7.2 7.2 Regression, cluster-robust standard errors (by division)

```
X_gap = np.column_stack([
    np.ones(len(events)),
    events["heat_anomaly_postfire"],
    events["log_severity"],
])
m = cluster_robust_ols(
    events["recovery_gap"],
    X_gap,
    events["ADM2_NAME"],
    ["Intercept", "heat_anomaly_postfire", "log_severity"],
)
m
```

	term	estimate	std_error_cluster	t_stat	p_value	n	clusters
0	Intercept	-8.119719	2.753318	-2.949068	0.006658	49	27
1	heat_anomaly_postfire	-0.010532	0.084046	-0.125309	0.901243	49	27
2	log_severity	0.788629	0.274892	2.868867	0.008072	49	27

7.3 7.3 Cluster bootstrap on the slope (robustness check)

Resampling whole divisions (not individual events) 2,000 times, to check the regression result isn't an artefact of one or two divisions with unusually many events.

```
rng = np.random.default_rng(42)
divs = events["ADM2_NAME"].unique()
boot_slopes = []
for _ in range(2000):
    sample_divs = rng.choice(divs, size=len(divs), replace=True)
```

```

boot_df = pd.concat([events[events.ADM2_NAME == d] for d in sample_divs], ignore_index=True)
boot_slopes.append(np.polyfit(boot_df["heat_anomaly_postfire"], boot_df["recovery_gap"], 1))
boot_slopes = np.array(boot_slopes)
ci_lo, ci_hi = np.percentile(boot_slopes, [2.5, 97.5])
print(f"bootstrap slope: mean={boot_slopes.mean():.4f}, 95% CI=[{ci_lo:.4f}, {ci_hi:.4f}]")
print(f"fraction of 2,000 resamples with a positive slope: {(boot_slopes > 0).mean():.3f}")

```

```

bootstrap slope: mean=0.0018, 95% CI=[-0.2006, 0.2312]
fraction of 2,000 resamples with a positive slope: 0.499

```

7.4 7.4 Kruskal-Wallis across heat terciles

```

kw_stat, kw_p = stats.kruskal(
    events[events.heat_tercile == "Low heat"]["recovery_gap"],
    events[events.heat_tercile == "Medium heat"]["recovery_gap"],
    events[events.heat_tercile == "High heat"]["recovery_gap"],
)
print(f"Kruskal-Wallis H={kw_stat:.3f}, p={kw_p:.4g}")

```

```

Kruskal-Wallis H=1.592, p=0.4511

```

7.5 7.5 Formal interaction test: is the heat-sensitivity actually *different* between the two metrics?

The regressions above test the gap variable directly. As a complementary cross-check, stack both recovery ratios into one long-format regression with a **metric x heat** interaction term – if that interaction is significant, it formally confirms NDVI and sightings respond differently to heat, without relying on the gap variable’s construction at all.

```

long_df = pd.concat([
    events.assign(metric="ndvi", log_ratio=events["log_ndvi_ratio"]),
    events.assign(metric="sightings", log_ratio=events["log_sight_ratio"]),
], ignore_index=True)
long_df["is_sightings"] = (long_df["metric"] == "sightings").astype(int)
long_df["heat_x_sightings"] = long_df["heat_anomaly_postfire"] * long_df["is_sightings"]
X_int = np.column_stack([
    np.ones(len(long_df)),
    long_df["heat_anomaly_postfire"],
])

```

```

    long_df["is_sightings"],
    long_df["heat_x_sightings"],
    long_df["log_severity"],
])
m_int = cluster_robust_ols(
    long_df["log_ratio"],
    X_int,
    long_df["ADM2_NAME"],
    ["Intercept", "heat_anomaly_postfire", "is_sightings", "heat_x_sightings", "log_severity"]
)
m_int

```

	term	estimate	std_error_cluster	t_stat	p_value	n	clusters
0	Intercept	2.449400	1.224498	2.000330	0.056010	98	27
1	heat_anomaly_postfire	-0.014240	0.036410	-0.391087	0.698919	98	27
2	is_sightings	-0.128835	0.496452	-0.259512	0.797284	98	27
3	heat_x_sightings	-0.011468	0.111097	-0.103227	0.918575	98	27
4	log_severity	-0.164598	0.112960	-1.457139	0.157051	98	27

=== Summary of all five tests ===

1. Spearman rank correlation: rho=0.001, p=0.9956
2. OLS (cluster-robust): coef=-0.0105, p=0.9012
3. Cluster bootstrap 95% CI: [-0.2006, 0.2312] (49.9% of resamples positive)
4. Kruskal-Wallis (heat terciles): H=1.592, p=0.4511
5. Metric x heat interaction: coef=-0.0115, p=0.9186

After aligning the heat window to the recovery window and measuring hot days against a 1991-2020 local normal, the heat effect is no longer supported. The Spearman correlation is essentially zero, the cluster-robust regression slope is near zero, the cluster bootstrap interval crosses zero widely, the heat tercile comparison is not significant, and the metric-by-heat interaction is also not significant. This is weaker than the earlier exploratory result, but it is the more defensible specification because the exposure window now matches the outcome window.

7.6 7.6 Assumption checks and interpretation

Test	Key assumptions	Checked / met here?	What it actually tells us
7.1 Spearman rank correlation	Monotonic (not necessarily linear) association; pairs independent of each other.	Monotonicity is a soft requirement Spearman handles by design. Independence is not met: the 49 events come from 27 divisions, and repeated fire events in the same division share local climate, soil, and species-baseline conditions – they are not independent draws. This test applies no correction for that clustering.	rho is approximately zero and p is approximately 0.996, so the rank test does not support a heat-gap relationship under this specification.
7.2 OLS, cluster-robust SEs	Linear relationship between predictors and the log-ratio outcome; no perfect multicollinearity; cluster-robust SEs are asymptotically valid, which in practice needs a reasonably large number of clusters.	Linearity in log-space is a defensible choice for ratio data. Only 27 clusters are available – below common rules of thumb for cluster-robust inference, so uncertainty should be read cautiously.	The heat coefficient is near zero and non-significant (p about 0.90), so this model does not support the hypothesis after controlling for fire severity.

Test	Key assumptions	Checked / met here?	What it actually tells us
7.3 Cluster bootstrap	Resampling whole divisions (not individual events) preserves within-division dependence; needs enough clusters for the resampled distribution to approximate the true one.	Correctly resamples at the division level. With only 27 divisions, the bootstrap distribution is still coarse.	The 95% bootstrap interval crosses zero widely and about half the resamples are positive, which is exactly what we would expect if the slope is not stable.
7.4 Kruskal-Wallis (terciles)	Independent observations across the 3 groups; roughly similar distribution shape across groups if the result is read as a median difference.	Independence is not fully met because events are clustered by division, and each tercile has only about 16 events.	H is small and p is about 0.45. The tercile comparison does not show a reliable shift in the recovery-gap distribution.
7.5 Metric x heat interaction	Same linearity/cluster-SE assumptions as 7.2, plus correlation between paired NDVI and sightings rows for the same fire event.	Division-clustering absorbs some repeated-division dependence but does not explicitly model event-level pairing.	The interaction term is near zero and non-significant (p about 0.92), so this complementary cross-check also fails to support differential heat sensitivity.

Bottom line on assumptions: the stricter method does not support the original heat-widens-the-gap hypothesis. The fairest interpretation is: **vegetation and wildlife recovery can diverge after fire, but this dataset does not provide robust evidence that the divergence increases with post-fire hot-day anomaly once the heat metric is aligned to the recovery window and benchmarked against a 1991-2020 local normal.**

8 8. Answering the research question

Is Australia's fire-affected environment doing well? The honest answer depends on which signal you trust:

- **If you trust satellite greenness alone: yes, mostly.** NDVI recovers to or above its pre-fire baseline within a few years after even the largest fires on record.
- **If you trust the wildlife record: more cautiously.** Koala and glider sighting-share recovery is noisier and more uncertain than NDVI recovery. It can lag behind vegetation in some events, but the stricter tests here do **not** show that this lag grows reliably with post-fire heat anomaly.

That divergence is the real finding, and it directly shapes the verdict: **Australia is not simply “doing fine” or “in crisis” – it has strong evidence of visible vegetation recovery, but weaker and more uncertain evidence for wildlife recovery.** A monitoring approach that leans on vegetation greenness alone risks missing habitat features that matter for arboreal mammals, such as hollows, mature canopy structure and connected forest. Given the small sightings-side sample ($n=49$, see Section 9), the responsible conclusion is a caution about monitoring, not a settled heat effect.

9 9. Limitations

- **Correlational, not causal;** heat, drought, and fire severity are entangled in Australian conditions and cannot be fully separated with the datasets in this folder.
- **Sightings are presence-only and effort-biased.** The share-of-year-total normalisation and the ≥ 200 -sightings habitat filter reduce but do not eliminate this; some years are dominated by single large survey campaigns in one division (see the companion EDA), which adds noise but should not by itself create a spurious *heat*-correlated pattern, since those campaigns aren’t targeted at hot years specifically.
- **The recovery ratio is a Relative Recovery Indicator (RRI)**, adapted from Frazier et al. (2018): $(\text{best post-fire value} - \text{fire-year value}) / (\text{pre-fire baseline} - \text{fire-year value})$. Their method uses a fixed $Y+4/Y+5$ window on precisely-dated per-pixel LandTrendr breakpoints from Landsat; here the “fire-year” value is just the annual division-wide NDVI/sightings composite for that calendar year, a much noisier proxy for the true disturbance trough, and the post-fire window is the best available of $Y+3/Y+4/Y+5$ (not fixed $Y+4/Y+5$) so that fires from 2019 onward – including Black Summer – remain in the NDVI data’s 2022 cutoff.
- **The disturbance-magnitude requirement (a measurable dip in the fire year) sharply reduces sample size**, especially for sightings: only 49 events retain a valid sightings-side RRI. This limits power and makes the estimated heat effect imprecise.
- **The heat metric is defensible but study-specific.** Hot days use a documented $\geq 35^\circ\text{C}$ threshold and the anomaly follows observed-minus-normal climatological practice, but applying that to an event-level recovery window is an analytical choice made for this study.
- **Three indicator species, fire-prone LGAs with cached data only** – this cannot be read as a claim about Australian ecosystems broadly.

10 10. Conclusion

The stronger version of the original claim does **not** survive the stricter method. After aligning the heat exposure window with the recovery window and using a documented 1991-2020 local hot-day normal, there is no robust evidence that post-fire heat anomaly widens the vegetation-wildlife recovery gap.

That is still a useful answer to “is Australia doing well?” The landscape’s visible vegetation recovery is real, but it should not be treated as complete evidence of wildlife habitat recovery. For forest-dependent arboreal mammals, the important recovery features include mature trees, hollows and connected canopy structure, not just greenness. The result is therefore a cautious environmental verdict: **Australia appears to be doing better on visible vegetation recovery than we can confidently say it is doing on wildlife recovery, and the available data are not yet strong enough to prove a heat-driven widening gap.**

11 11. Appendix A: Feature Definitions

Term	Definition
Major fire event	A division-year with at least one recorded fire $\geq 10,000$ ha – separates landscape-scale wildfires from routine hazard-reduction burns, which are recorded in almost every year in most fire-prone divisions.
Pre-fire baseline	The mean of the two years immediately before the fire year (Y-1, Y-2), computed separately for NDVI and for sightings share.
Fire-year value (Y0)	The NDVI/sightings-share value in the fire year itself – stands in for the “disturbance trough” that Frazier et al. (2018) locate precisely from per-pixel LandTrendr segmentation; here it is just the annual division-wide composite for that calendar year, a noisier proxy for the same idea (see Section 9).
Post-fire recovery window	The best available value among Y+3, Y+4, Y+5 – adapted from the paper’s fixed Y+4/Y+5 window so that fires from 2019 onward (including Black Summer) are not dropped by the NDVI data’s 2022 cutoff.

Term	Definition
NDVI/sightings recovery ratio (RRI)	Relative Recovery Indicator, adapted from Frazier et al. (2018): $RRI = \frac{ARI}{\text{disturbance magnitude}}$ $\text{disturbance magnitude} = (\max(\text{NDVI over the recovery window}) - \text{NDVI}(Y_0)) / (\text{mean}(\text{NDVI}(Y-1), \text{NDVI}(Y-2)) - \text{NDVI}(Y_0))$. Same construction for sightings share, with $\text{eps} = 1\text{e-}6$ added throughout to avoid division by zero. ~ 1.0 = recovered back to the pre-fire baseline relative to how hard the fire year was hit; below 1 = still short of that; above 1 = overshot it. Only defined (and only computed) when the disturbance magnitude is positive – see Section 9.
Sightings share	A division’s koala/glider sightings in a given year, divided by total sightings across all fire-prone divisions that year – a partial correction for the nationwide growth in citizen-science reporting effort over time.
Heat anomaly (post-fire)	$\text{mean}(\text{hot days} \geq 35C \text{ over the same available recovery years used for RRI, among } Y+3/Y+4/Y+5) - \text{mean}(\text{hot days} \geq 35C \text{ over } 1991\text{-}2020 \text{ for that division})$. Hot days follow fixed-threshold hot-day index practice; the 1991-2020 baseline follows climatological-normal practice. The metric captures “unusually hot for this place” rather than just “a hot place.”
Recovery gap	$\log(\text{NDVI recovery ratio}) - \log(\text{sightings recovery ratio})$. Zero = vegetation and wildlife recovered by the same proportional amount; positive = vegetation recovered relatively better than wildlife.
Fire severity (log_severity)	$\log(\text{maximum single fire area in hectares recorded in that division-year}) - \log\text{-transformed because fire size spans several orders of magnitude. Included as a regression control variable.}$

Term	Definition
Habitat division filter	Only divisions with ≥ 200 total koala/glider sightings across the whole record are included in sightings-based tests, to exclude LGAs (mostly inland cropping/rangeland) where these species barely occur and the sightings ratio would just be noise.
Heat tercile	Events split into three equal-sized groups (Low / Medium / High) by post-fire heat anomaly, used for the non-parametric comparisons in Sections 6.2 and 7.4.

12 12. References

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